

Chan Yat Kwan

simontlchanyk@gmail.com • ☎ 66002881 • <https://simontlcyk.github.io>

EDUCATION

The University of Hong Kong

Bachelor of Engineering in Computer Science | CGPA: 3.60

June 2026

Relevant Coursework: Data Structures and Algorithms, Database, Machine Learning, Cyber Security, OOP, Game Development, Mobile App Development, World Wide Web, Communication Network, Computer Organization, Software Engineering, Data Science

WORK EXPERIENCE

XR + AI Developer Intern | **VOTANIC Limited** | *Unity, C#, Docker, Node.js, Express.js*

June 2025 - August 2025

Develop XR and AI solution with XR platform integration

- Developed demo project in **Unity** in a team of 2, scripted in **C#**, utilized **machine learning** framework for **face detection**, to implement **off-axis projection** via user head tracking.
- Integrated a **photo-to-3D model generation** pipeline into a web-based 3D model editing application, using **Docker**, **Node.js**, and **Express.js** for containerization and deployment.

STEM Intern | **Up Innovations Limited** | *Unity, C#, AWS, Firebase*

July 2024 - August 2024

Develop interactive applications for Special Educational Needs

- Designed and built a game prototype in **Unity**, scripted in **C#**, utilizing **machine learning** framework for **facial gesture recognition**, which helps users do oral motor exercise.
- Assisted in implementing **text-to-speech** and **speech-to-text** functions in **Unity** with HuggingFace API and **AWS lambda**.

PROJECTS

[Video Game with Optimization & AI Player](#)  | *Unity, C#, Firebase*

Worked individually to make a 2D action game with performance optimization and training of AI Player

- Designed and developed game in **Unity**, scripted in **C#**, with **Firebase** for **authentication** and **cloud data save-load**.
- Utilized **state design pattern** to implement game character behaviors, to improve maintainability and scalability of behavior system.
- Implemented **parallel processing** that increases game frame rate by ~90%; and implemented **object pooling** that reduces high frame time occurrence and avoids game stuttering.
- Trained **AI models** that simulate player behavior in playing the game via **reinforcement learning** and **imitation learning**.

[Online Library Catalogue](#)  | *HTML, JavaScript, CSS, Firebase*

Worked individually to make a book searching website for an organization

- Designed and built a static website with **HTML**, **JavaScript**, **CSS** that allows users to search for information and status of books from the organization's library, with custom search criteria and sorting on search results.
- Utilized **Firebase** for **cloud data storage** of book records and **authorization** of record upload and update by members.

[PlanSmart](#)  | *Android Studio, Kotlin, Firebase*

Worked in a team of 2 to make a mobile app for scheduling and planning

- Collaboratively designed and built mobile app in **Android Studio**, scripted in **Kotlin**.
- Implemented functions that allow users to add, edit, and view items and fields information.
- Utilized **Firebase** for **authentication** and **cloud data storage**, allowing user data synchronization on multiple devices.

TECHNICAL SKILLS

Language: C#, JavaScript, HTML, CSS, Python, Java, C++, Kotlin, R, SQL

Technical Framework: Node.js, Express.js

Technical Tools / Software: Unity, Git, Visual Studio, VS Code, Docker, Firebase, AWS, Android Studio

LANGUAGE

Cantonese - Native

English - Fluent

Mandarin - Fluent